

EIGENANALYSIS

Given $A \in \mathbb{C}^{n \times n}$, if a vector v ($v \neq 0$)
and a scalar λ satisfies

$$Av = \lambda \cdot v.$$

then λ is called an eigenvalue of A
and v is called an eigenvector of A
corresponding to the eigenvalue λ .

(λ, v) is called an eigenpair of A .

$$Av = \lambda v \Leftrightarrow (A - \lambda I)v = 0$$

$$\Leftrightarrow A - \lambda I \text{ is}$$

rank-deficient -

$$\Leftrightarrow N(A - \lambda I) \neq \{0\}$$

Reading Assignment: Read about determinant.

$$N(A - \lambda I) \neq \{0\}$$

$$\Leftrightarrow \text{rank}(A - \lambda I) < n$$

$$\Leftrightarrow \underline{\det(A - \lambda I) = 0}$$

$\det(A - \lambda I)$ is a polynomial in λ .

$$\det(A - \lambda I) = c_0 + c_1 \lambda + c_2 \lambda^2 + \dots + c_n \lambda^n$$

$$c_0 + c_1 \lambda + c_2 \lambda^2 + \dots + c_n \lambda^n = 0$$

λ is an eigenvalue of A if and only if it is a root of the following polynomial:-

$$f_A(\lambda) = \det(A - \lambda I)$$

(characteristic polynomial of A).

$f_A(\lambda)$ has n (not necessarily distinct) roots.

Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be the distinct roots of $f_A(\lambda)$, and $\alpha_1, \alpha_2, \dots, \alpha_k$ be their multiplicities.

$$f_A(\lambda) = (\lambda - \lambda_1)^{\alpha_1} (\lambda - \lambda_2)^{\alpha_2} \cdots (\lambda - \lambda_k)^{\alpha_k}$$

$$\sum_{i=1}^k \alpha_i = n$$

α_i : algebraic multiplicity of the eigenvalue λ_i

Every $A \in \mathbb{C}^{n \times n}$ has n eigenvalues (including repetitions)

Eigenspace

Suppose $\bar{\lambda}$ is an eigenvalue of A .

Is there a $\bar{v}$ s.t. $A\bar{v} = \bar{\lambda}\bar{v}$?

Yes $\bar{v} \in N(A - \bar{\lambda}I)$
 $(x \neq 0)$

If $x \in N(A - \bar{\lambda}I) \Rightarrow x$ is
 an eigenvector $\hat{x}$ of A corresp.
 to eigenvalue $\bar{\lambda}$.

$$E_{\bar{\lambda}} = \{v, v \neq 0, \text{ s.t. } Av = \bar{\lambda}v\}$$

$$E_{\bar{\lambda}} \cup \{0\} = N(A - \bar{\lambda}I)$$

$\dim(N(A - \bar{\lambda}I))$ is called the
 geometric multiplicity of $\bar{\lambda}$,
 denoted by $g_A(\bar{\lambda})$

If $v_1, v_2 \in E_{\bar{\lambda}}$

then $Av_1 = \bar{\lambda}v_1$

$$Av_2 = \bar{\lambda}v_2$$

$A(v_1 + v_2) = \bar{\lambda}(v_1 + v_2) \Rightarrow v_1 + v_2$ is also an eigenvector

Similarly for $\alpha \neq 0$,

$$\begin{aligned} A(\alpha v_1) &= \alpha \cdot Av_1 \\ &= \alpha \cdot \bar{\lambda}v_1 \\ &= \bar{\lambda}(\alpha v_1) \end{aligned}$$

$\Rightarrow \alpha v_1$ is also an eigenvector

Fact

Let $\alpha_A(\bar{\lambda})$ and $g_A(\bar{\lambda})$ be the

algebraic & geometric multiplicities
of $\bar{\lambda}$

then it holds

$$1 \leq g_A(\bar{\lambda}) \leq \alpha_A(\bar{\lambda})$$

If $\bar{\lambda}$ repeats "once" as a root
of $f_A(\lambda)$ then

$$\alpha_A(\bar{\lambda}) = 1$$

$$\Rightarrow g_A(\bar{\lambda}) = 1$$

Application

1. DSP (Later...)

2. Linear differential equations with
constant coefficients.

$$\frac{dx(t)}{dt} = \alpha x(t), \quad x(0) = C_0$$

solution is

$$x(t) = C_0 e^{\alpha t} = C_0 \underbrace{e^{\alpha t}}_{\text{e}^{j\gamma t}} \cdot \underbrace{e^{j\gamma t}}_{\text{e}^{j\gamma t}}$$

$$\rightarrow C_0 e^{\alpha t} (\cos(\gamma t) + j \sin(\gamma t))$$

$$\alpha = \lambda + j\gamma$$

$$\begin{bmatrix} \frac{dx_1(t)}{dt} \\ \frac{dx_2(t)}{dt} \\ \vdots \\ \frac{dx_n(t)}{dt} \end{bmatrix} = \begin{bmatrix} \sum_{j=1}^n A_{1j} x_j(t) \\ \sum_{j=1}^n A_{2j} x_j(t) \\ \vdots \\ \sum_{j=1}^n A_{nj} x_j(t) \end{bmatrix}$$

Solve for
 $x_1(t), \dots$
 $x_n(t)$

$$x_1(0) = C_1, \quad x_2(0) = C_2, \dots, \quad x_n(0) = C_n$$

How do you solve for $x_1(t), \dots, x_n(t)$
 using the solution for $n=1$?

Topic: Diagonalization -

SIMILAR MATRICES

Two matrices $A, B \in \mathbb{C}^{n \times n}$ are said to be similar if there exists a full rank matrix $P \in \mathbb{C}^{n \times n}$ s.t.

$$A = \underbrace{P^{-1}}_Q B P, \quad A = Q B Q^{-1}$$

Eigenvalues of similar matrices?

λ is an eigenvalue of A

$$\Leftrightarrow \det(A - \lambda I) = 0$$

$$\Leftrightarrow \det(P^{-1} B P - \lambda I) = 0$$

$$\Leftrightarrow \det(P^{-1} (B - \lambda I) P) = 0$$

$$\Leftrightarrow \cancel{\det(P^{-1})} \det(B - \lambda I) \cancel{\det(P)} = 0$$

$$\Leftrightarrow \det(B - \lambda I) = 0$$

$$\begin{aligned} \det(P^{-1}) &= \frac{1}{\det(P)} \\ &\quad \left(P \text{ is non-singular} \right) \end{aligned}$$

Similar matrices share same eigenvalues.

DIAGONALIZATION

A matrix $A \in \mathbb{C}^{n \times n}$ is said to be diagonalizable if it is similar to a diagonal matrix.

ie $A = P\Lambda P^{-1}$, where Λ is some diagonal matrix ($n \times n$)

Note: If A is diagonalizable

$$P = [p_1 \ p_2 \ \dots \ p_n]$$

$$\Lambda = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{bmatrix}$$

$$\Rightarrow AP = P\Lambda$$

$$\Rightarrow Ap_1 = p_1 \lambda_1 = \lambda_1 p_1$$

$$Ap_2 = \lambda_2 p_2$$

$$\vdots$$

$$Ap_n = \lambda_n p_n$$

$\Rightarrow (\lambda_i, p_i)$ is an eigenpair of A .

→ A has n linearly independent eigenvectors

(in general, this may not be true)

Converse is also true:

If $A \in \mathbb{C}^{n \times n}$ has n linearly independent eigenvectors then A is similar to a diagonal matrix. Suppose $v_1, \dots, v_n$ are the lin. indep. eig. of A & let $\lambda_1, \dots, \lambda_n$ be the corresponding eigenvalues

$$Av_1 = \lambda_1 v_1$$

$$Av_2 = \lambda_2 v_2$$

$\vdots$

$$Av_n = \lambda_n v_n$$

$$A \underbrace{[v_1 \dots v_n]}_{\text{rank} = n} = \underbrace{[v_1 \ v_2 \dots v_n]}_V \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

$$\Rightarrow \boxed{A = V \Lambda V^{-1}}$$

A matrix A is diagonalizable if and only if it has n linearly independent eigenvectors.

Q ∴ What is the corresponding necessary & sufficient condition in terms of eigenvalues of A ?

What happens if A is Diagonalizable?

$$\frac{dx_1(t)}{dt} = \sum_{j=1}^n A_{1j} x_j(t)$$

$\vdots$

$$\frac{dx_n(t)}{dt} = \sum_{j=1}^n A_{nj} x_j(t)$$

$$\begin{bmatrix} \frac{dx_1(t)}{dt} \\ \vdots \\ \frac{dx_n(t)}{dt} \end{bmatrix} = A \cdot \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$$

$x(t)$

Suppose A is diagonalizable

$$\Rightarrow A = P \Lambda P^{-1}$$

$$\frac{dx(t)}{dt} = A x(t) = P \Lambda \underline{P^{-1} x(t)}$$

$$\text{Define } y(t) = P^{-1} x(t)$$

$$\frac{dx(t)}{dt} = P \Lambda y(t)$$

$$P^{-1} \frac{dx(t)}{dt} = \Lambda y(t)$$

$$\frac{d}{dt} P^{-1} x(t) = \Lambda y(t)$$

$$\frac{dy(t)}{dt} = \Lambda y(t)$$

$$\frac{dy_1(t)}{dt} = \lambda_1 y_1(t) \rightarrow y_1(t) = e^{\lambda_1 t} c'_1$$

$$\frac{dy_2(t)}{dt} = \lambda_2 y_2(t)$$

$$\frac{dy_n(t)}{dt} = \lambda_n y_n(t)$$

$$y_n(t) = e^{\lambda_n t} c'_n$$

$$\begin{bmatrix} y_1(t) \\ \vdots \\ y_n(t) \end{bmatrix} = P^{-1} x(t)$$

$$x(t) = P \cdot \begin{bmatrix} y_1(t) \\ \vdots \\ y_n(t) \end{bmatrix}$$

$$\Rightarrow \boxed{x_i(t) = \sum_{k=1}^n \alpha_{ik} e^{\lambda_k t}}$$

$\Rightarrow$ If $\operatorname{Re}(\lambda_k) \geq 0$ for some k
(and $\alpha_{ik} \neq 0$), then

Let $x_i(t) \rightarrow +\infty$ (if $\alpha_{ik} > 0$)
 $t \rightarrow \infty$

— unstable system.

If $\operatorname{Re}(\lambda_k) < 0$ for all k

then $\lim_{t \rightarrow \infty} x_i(t) = 0$ } $\rightarrow$ stable system

Hermitian Symmetric Matrices

Not all square matrices are diagonalizable.

A matrix $A \in \mathbb{C}^{n \times n}$ is Hermitian if

$$A = A^H \quad \left((A^H)_{ij} = A_{ji}^* \right)$$

Quick observation:

All eigenvalues of a Hermitian matrix are real.

Suppose (λ, v) is an eigenpair

$$Av = \lambda v.$$

$$v^H A^H = \lambda^* v^H$$

$$\Rightarrow \underbrace{v^H A^H}_{=A} v = \lambda^* \|v\|_2^2 \quad \text{--- (1)}$$

$$\Rightarrow v^H Av = \lambda v^H v = \lambda \|v\|_2^2 \quad \text{--- (2)}$$

$$\Rightarrow \lambda^* \|v\|_2^2 = \lambda \|v\|_2^2 \quad (\|v\|_2 \neq 0)$$

$\Rightarrow \lambda$ is real.

Diagonalization of Hermitian Matrices

If A is Hermitian, then A permits the following factorization

$$A = \underset{(n \times n)}{U} \underset{(n \times n)}{\Lambda} \underset{(n \times n)}{U}^H$$

$\Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$, U is a unitary matrix

$$\Rightarrow U^H U = I \Rightarrow U^{-1} = U^H$$

Does this also imply

$$U U^H = I \quad ? \text{ Yes}$$

$$\left(\text{Since } U U^H = U U^{-1} = I \right)$$

$$A = U \Lambda U^{-1}$$

$\Rightarrow$ columns of U are eigenvectors of A
and $\lambda_1, \dots, \lambda_n$ are eigenvalues of A .

$\Rightarrow$ Since eigenvalues of Hermitian A are real, they can be ordered.

$$\Rightarrow \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$$

(eigenvalues can be negative).

Rayleigh Quotient

Suppose $A = A^H \in \mathbb{C}^{n \times n}$

Rayleigh Quotient is defined as a function;

$$R_A: \mathbb{C}^n \setminus \{0\} \rightarrow \mathbb{R}$$

$$R_A(x) = \frac{x^H A x}{x^H x}$$

$$(x \in \mathbb{C}^n)$$

Why is $R_A(x)$ real-valued?

$R_A(x)$ is real if

$$R_A(x) = R_A^*(x) = R_A^H(x)$$

Notice

$$R_A^H(x) = \frac{x^H \overset{\sim A}{A^H} x}{x^H x} = \frac{x^H A x}{x^H x} = R_A(x)$$

Claim:

$$\lambda_1 \geq R_A(x) \geq \lambda_n$$

Proof:

$$R_A(x) = \frac{x^H A x}{x^H x}$$

$$A = U \Lambda U^H$$

$$= \frac{x^H U \Lambda U^H x}{x^H x}$$

$y = U^H x$
 $y \neq 0$ since
 U is full rank
and $x \neq 0$

$$= \frac{y^H \Lambda y}{y^H y}$$

$$[y^H y = x^H U^H U x = x^H x]$$

$$= \frac{[y_1^* \dots y_n^*] \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}}{y^H y}$$

$$= \frac{\sum_{i=1}^n \lambda_i |y_i|^2}{y^H y} = \frac{\lambda_1 |y_1|^2 + \dots + \lambda_n |y_n|^2}{y^H y}$$

Note $\lambda_i \geq \lambda_n, \quad i=1, 2, \dots, n-1$

$$\lambda_i |y_i|^2 \geq \lambda_n |y_i|^2$$

(since $|y_i|^2 \geq 0$)

$$R_A(x) = \frac{\sum_{i=1}^n \lambda_i |y_i|^2}{y^H y}$$

$$\geq \frac{\sum_{i=1}^n \lambda_n |y_i|^2}{y^H y}$$

$$= \lambda_n \frac{\sum_{i=1}^n |y_i|^2}{y^H y}$$

$$= \lambda_n$$

$$R_A(x) \geq \lambda_n, \quad x \neq 0$$

Same idea to show $R_A(x) \leq \lambda_1$
 $x \neq 0$

When is

$$R_A(x) = \lambda_1 \quad ?$$

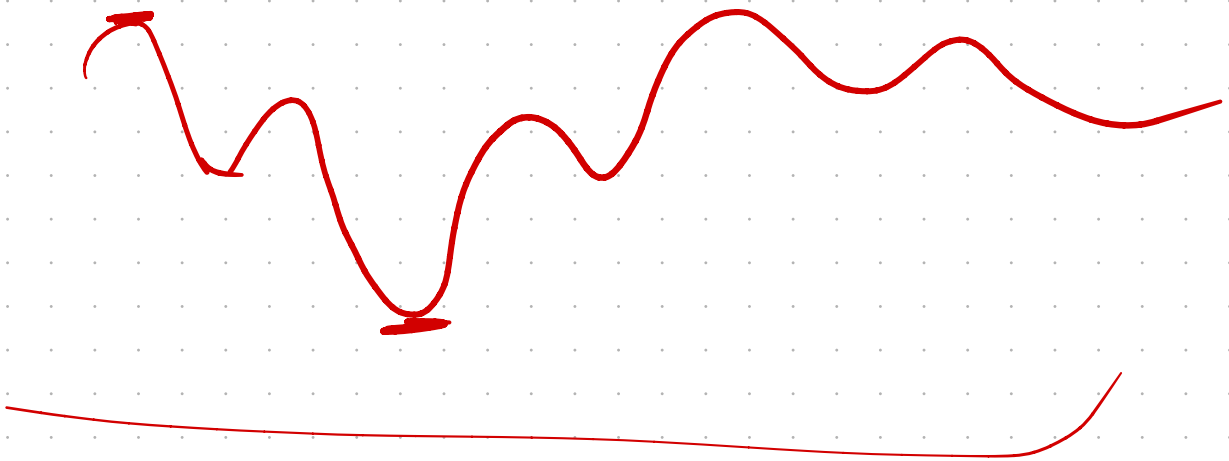
Recall

$$A = [v_1 \dots v_n] \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \begin{bmatrix} v_1^H \\ \vdots \\ v_n^H \end{bmatrix}$$

$$Av_1 = \lambda_1 v_1$$

$$R(v_1) = \frac{v_1^H \underbrace{Av_1}_{\lambda_1 v_1}}{\underbrace{v_1^H v_1}_{=1}} = \lambda_1 \frac{v_1^H v_1}{v_1^H v_1} = \lambda_1$$

$$R(u_n) = \lambda_n.$$



Hermitian Symmetric Positive Semidefinite Matrices (SPSD)

A matrix A , ($A = A^H$) is said to be PSD if

$$x^H A x \geq 0, \quad \forall x$$

A is PD if

$$x^H A x > 0, \quad \forall x \neq 0.$$

A is Hermitian

$$A = U \Lambda U^H$$

Fact:

All eigenvalues of a SPSD Matrix A are nonnegative.

Suppose (λ, v) is an eigenpair

$$v^H A v \underset{\lambda v}{=} \lambda v^H v \geq 0$$

$$\Rightarrow \lambda \|v\|_2^2 \geq 0 \Rightarrow \lambda \geq 0$$

All eigenvalues of a SPD matrix A are positive

(same proof as above)

rank of A if A is SPD?

$$A = U \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} U^H$$

$$\text{rank}(A) = \text{rank} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} = n \quad \text{if} \quad \lambda_1 \geq \dots \geq \lambda_n \geq 0$$

If A is SPD

$$A = U \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} U^H$$

$$\lambda_i > 0 \quad \forall i$$

$$\lambda_i = \underbrace{\sqrt{\lambda_i}} \underbrace{(\sqrt{\lambda_i})}$$

$$\text{Let } \beta_i = \sqrt{\lambda_i} > 0$$

$$\Rightarrow A = U \begin{bmatrix} \sqrt{\lambda_1} & & \\ & \ddots & \\ & & \sqrt{\lambda_n} \end{bmatrix} \cdot \begin{bmatrix} \sqrt{\lambda_1} & & \\ & \ddots & \\ & & \sqrt{\lambda_n} \end{bmatrix} U^H$$

$$= \underbrace{U \begin{bmatrix} \sqrt{\lambda_1} & & \\ & \ddots & \\ & & \sqrt{\lambda_n} \end{bmatrix}}_B \underbrace{U^H U}_{I} \begin{bmatrix} \sqrt{\lambda_1} & & \\ & \ddots & \\ & & \sqrt{\lambda_n} \end{bmatrix} U^H$$

$$= B \cdot B$$

B is "square-root" of A .

B is also SPD.

For PSD matrices,

$x^H A x$ can be zero for
some x .

(ie a PSD matrix can have zero
eigenvalues).

If A has zero eigenvalue,
what is the corresponding
eigenspace?

$$\Rightarrow \mathcal{N}(A)$$

Random Vectors

$X \in \mathbb{R}^n$ is a random vector

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad x_1, \dots, x_n \text{ are random variables}$$

$$E(XX^T) = R_{xx} \quad (\text{correlation matrix})$$

XX^T is rank 1

But R_{xx} can have rank "n".

Claim:

R_{xx} is a SPSD matrix.

Proof:

$$V^T R_{xx} V = V^T E(XX^T) V$$

$$= E\left(v^T x \underbrace{x^T v}_u\right) = E(u^2) \geq 0$$

$\Rightarrow R_{xx}$ is SPSD.

Covariance matrix:

$$E\left([x - E(x)][x - E(x)]^T\right)$$

= Covariance matrix.
